

Multiple Choice (10 marks)

1. The quantum efficiency of a photon detector is 0.1. If 100 photons are sent into the detector, one after the other, the detector will detect photons
 - (a) exactly 10 times
 - (b) an average of 10 times, with an rms deviation of about 0.1
 - (c) an average of 10 times, with an rms deviation of about 1
 - (d) an average of 10 times, with an rms deviation of about 3
2. Which of the following ions CANNOT be used as a dopant in germanium to make an n-type semiconductor?
 - (a) As
 - (b) P
 - (c) Sb
 - (d) B
3. A satellite of mass m orbits a planet of mass M in a circular orbit of radius R . The time required for one revolution is
 - (a) independent of M
 - (b) proportional to $m^{(1/2)}$
 - (c) linear in R
 - (d) proportional to $R^{(3/2)}$
4. A three-dimensional harmonic oscillator is in thermal equilibrium with a temperature reservoir at temperature T . The average total energy of the oscillator is
 - (a) $(1/2) k T$
 - (b) kT
 - (c) $(3/2) k T$
 - (d) $3 kT$
5. By definition, the electric displacement current through a surface S is proportional to the
 - (a) rate of change of the electric flux through S
 - (b) electric flux through S
 - (c) time integral of the magnetic flux through S
 - (d) rate of change of the magnetic flux through S

True / False (10 marks)

Answer True or False

6. True or False: If a photon detector has a quantum efficiency of 0.1, it will detect an average of 10 photons with an rms deviation of about 4.
7. True or False: Electromagnetic radiation emitted from a nucleus is most commonly in the form of gamma rays.
8. True or False: An observer must move at $0.60c$ parallel to a 1.00 m rod to measure its length as 0.80 m due to length contraction.
9. True or False: The electric displacement current through a surface S is proportional to the rate of change of the magnetic flux through S .
10. True or False: The frequency of the next higher harmonic for a closed organ pipe with a fundamental frequency of C (131 Hz) is 393 Hz .

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the process of ionization in helium, explaining the energy dynamics involved. Specifically, analyze how the total energy required to remove both electrons relates to the energy needed to remove a single electron.
12. A tube of water moves at half the speed of light relative to a lab frame. Analyze the speed of a light beam entering the tube in the same direction, considering the index of refraction of water.

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